

Exercise Sheet 5

Data as a Signal versus Noise

Use of AI You may use AI tools to help draft code, but your final answers should show your own judgment and explain why the results look like that.

Setup in Python Use `numpy`, `pandas`, and `matplotlib`. For a real symmetric matrix, `np.linalg.eigvalsh(S)` returns eigenvalues in ascending order. Use `np.linalg.eigh(S)` when eigenvectors are needed.

Notation Let the following to be a return matrix: $\mathbf{R} \in \mathbb{R}^{n \times m}$. Center each row to obtain $\mathbf{X}$, and use divisor m : $\hat{\mathbf{C}} = \mathbf{X}\mathbf{X}^\top/m$. Order eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$. For iid noise of variance σ^2 , write $\gamma = n/m$, $a = \sigma^2(1 - \sqrt{\gamma})^2$, $b = \sigma^2(1 + \sqrt{\gamma})^2$. The Marchenko–Pastur (MP) continuous density is given by

$$f_{\gamma, \sigma^2}(x) = \frac{\sqrt{(b-x)(x-a)}}{2\pi\gamma\sigma^2 x} \mathbf{1}_{\{a < x < b\}},$$

with an atom of mass $(1 - 1/\gamma)_+$ at zero.

Problem 1

- a) Classify the following as a Wigner matrix or not, with a reason.
- i) A 3×3 matrix has zero diagonal and three independent signs $\xi_{12}, \xi_{13}, \xi_{23} \in \{-1, +1\}$, each sign equally likely, mirrored below the diagonal.
 - ii) A 3×3 matrix has zero diagonal and the same random sign $\xi \in \{-1, +1\}$ in every off-diagonal position.
- b) One realization of construction i) is

$$\mathbf{W} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Find the eigenvalues of $\mathbf{W}/\sqrt{3}$ by hand and write its empirical spectral measure.

- c) Let $\mathbf{X} \in \mathbb{R}^{n \times m}$ and $\mathbf{S} = \mathbf{X}\mathbf{X}^\top/m$. For $n = 400$ and $m = 200$, how many zero eigenvalues must $\mathbf{S}$ have if $\mathbf{X}$ has rank 200?
- d) Calculate the MP support $a = \sigma^2(1 - \sqrt{\gamma})^2$, and $b = \sigma^2(1 + \sqrt{\gamma})^2$ and the mass at zero for $(\gamma, \sigma^2) = (2, 1)$.

Problem 2

- Generate a zero-diagonal Wigner matrix with $n = 500$ and independent $N(0, 1)$ upper entries. Mirror the upper triangle, divide by $\sqrt{n}$, and plot the two eigenvalue histograms beside the semicircle density $\rho(x) = \frac{1}{2\pi} \sqrt{4 - x^2} \mathbf{1}_{\{|x| \leq 2\}}$.
- Generate an iid Gaussian matrix $\mathbf{X}$ with known zero mean and variance one for $(n, m) = (400, 200)$. Estimate $\mathbf{S} = \frac{1}{m} \mathbf{X} \mathbf{X}^\top$ and plot its eigenvalues together with the correctly scaled MP density. Mark the edges a and b . Analyze the number of zero eigenvalues and explain why the semicircle law is not appropriate for analyzing these matrices.
- The code below is intended for variables in rows, dates in columns, and iid Gaussian noise of variance 4. Identify and correct every line that is incompatible with this interpretation. State the dimension of the corrected covariance matrix and the correct value of γ .

```
n, m = 120, 240
R = 2 * rng.normal(size=(n, m))
X = R - R.mean(axis=0, keepdims=True)
S = X.T @ X / n
eig = np.linalg.eigvals(S) / np.sqrt(n)
gamma = m / n
a, b = (1 - np.sqrt(gamma))*2, (1 + np.sqrt(gamma))*2
```

Problem 3 Continue with the same financial data as Sheet 3, Problem 3 for SPY, QQQ, TLT, and HYG, but use the period 2018–2019. Store assets in rows and dates in columns.

- Standardize each asset using its sample mean and standard deviation over this period, and compute the 4×4 sample correlation matrix $\hat{\mathbf{C}}$.
- Calculate the unit noise MP upper edge using $\gamma = 4/m$ and add it as a reference line to a scree plot. Explain why this line should be treated as a heuristic. Mention at least three assumptions of the iid MP noise model that might not hold for daily financial returns.